

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

SPECTRAL DECOMPOSITION

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Spectral Decomposition

Fact: Let M be any real symmetric $d \times d$ matrix. Then M has orthonormal eigenvectors $u_1, \dots, u_d \in \mathbb{R}^d$ and corresponding eigenvalues $\lambda_1, \dots, \lambda_d$.

Spectral decomposition: Another way to write M :

$$M = \underbrace{\begin{pmatrix} \uparrow & \uparrow & \uparrow & \uparrow \\ u_1 & u_2 & \cdots & u_d \\ \downarrow & \downarrow & \downarrow & \downarrow \end{pmatrix}}_{U: \text{ columns are eigenvectors}} \underbrace{\begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_d \end{pmatrix}}_{\Lambda: \text{ eigenvalues on diagonal}} \underbrace{\begin{pmatrix} \leftarrow u_1 \rightarrow \\ \leftarrow u_2 \rightarrow \\ \leftarrow \vdots \rightarrow \\ \leftarrow u_d \rightarrow \end{pmatrix}}_{U^T}$$

Thus $Mx = U\Lambda U^T x$:

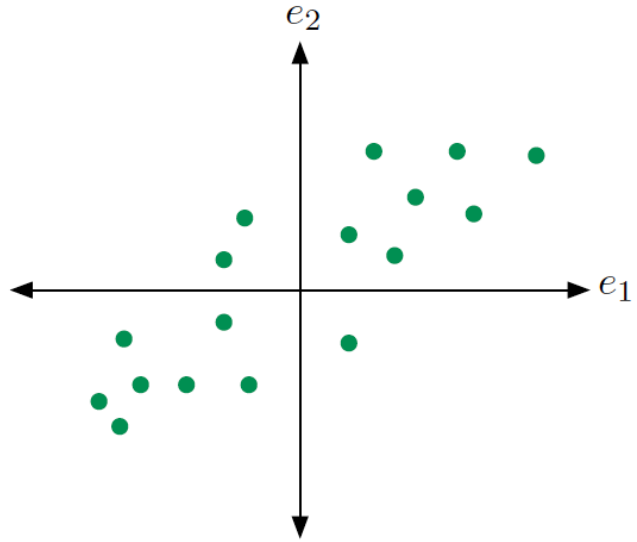
- U^T rewrites x in the $\{u_i\}$ coordinate system
- Λ is a simple coordinate scaling in that basis
- U sends the scaled vector back into the usual coordinate basis

Apply spectral decomposition to the matrix we saw earlier:

$$M = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$$

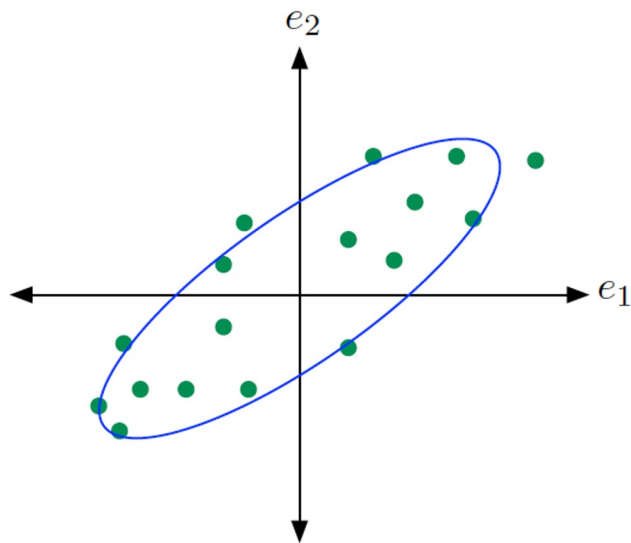
- Eigenvectors $u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$
- Eigenvalues $\lambda_1 = -1$, $\lambda_2 = 3$.

Principal Component Analysis Revisited



Data vectors $X \in \mathbb{R}^d$

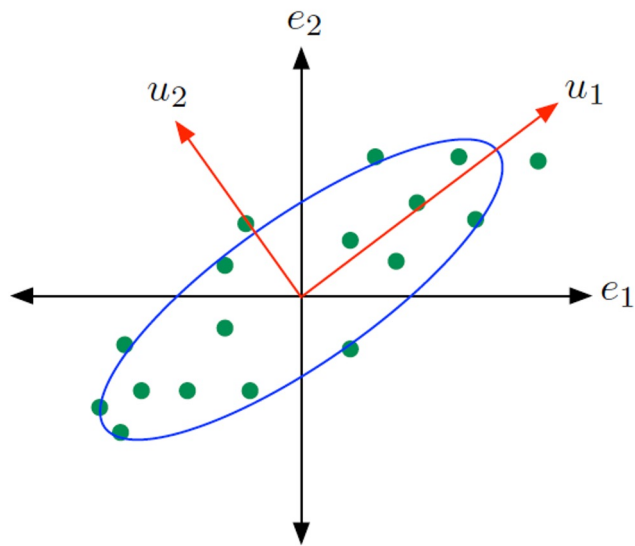
Principal Component Analysis Revisited



Data vectors $X \in \mathbb{R}^d$

- $d \times d$ covariance matrix Σ is symmetric.
- Eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$
- Eigenvectors $u_1, \dots, u_d$.

Principal Component Analysis Revisited



Data vectors $X \in \mathbb{R}^d$

- $d \times d$ covariance matrix Σ is symmetric.
- Eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$
- Eigenvectors $u_1, \dots, u_d$.
- $u_1, \dots, u_d$: another basis for data.
- Variance of X in direction u_i is λ_i .
- Projection to k dimensions:
 $x \mapsto (x \cdot u_1, \dots, x \cdot u_k)$.

What is the covariance of the projected data?